

Tracking Data Message Files

The Tracking Data Message (TDM) format is used for delivering tracking data to the user. The TDM format is defined by the [CCSDS 503.0-B-2](#) standard. This section defines the specifics of how TDM files are generated, and the TDM features that are used by the KSAT Lunar system.

For a given track operation, up to 4 different TDM files may be generated.

- Radiometric tracking data (range, carrier phase, range rate) is stored in a TRACK file.
- Antenna pointing angles (azimuth, elevation) are stored in an ANGLE file.
- Signal metrics (receive power, SNR, etc) are stored in a SIGMET file.
- Meteorology (temperature, pressure) are stored in a METEO file.

The generation of these files depends on the capabilities present at a given site. For certain sites, some capability may not be present, in which case the associated TDM file will not be generated. For example, some sites may not have access to weather data, or the antenna controller may not provide angle readback. In these cases, the METEO file and ANGLE file, respectively, would not be generated at these sites.

The current TDM file structure is only designed for supporting two-way ranging measurements between a single ground antenna and a single spacecraft. Extensions to the TDM definition in this document will be required to support more advanced concepts, such as VLBI, relay satellites, etc.

File Name Convention

The file name for the TDM files follows a convention so that the contents and relevance of the file can easily be identified from the file name itself, without having to examine the contents.

<TYPE>_<GSID>_<SVID>_<DATE>.tdm

The portions of the file name are:

- TYPE – designates the contents of the TDM file. The type designator is defined for each file type.
- GSID – ground station ID, the KSAT-unique ID for the ground station.

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SVID – the spacecraft ID, usually the COSPAR ID.

DATE – the date and time of creation of the file. This should be the same value, and the same format, as the creation date placed in the file header, except that the colons (':') in the time field should be replaced with dashes ('-').

.tdm – extension indicating TDM file format.

For maximum cross-platform support, the filename shall only contain the following characters:

- Alphanumeric characters from the ASCII character set ([A-Za-z0-9]).
- Dash (-) and underscore (_) are allowed, except they cannot be used at the beginning of the file name.
- Period (.) should only be used to separate the file name from the extension.

TDM Header

The header provides high-level information about the TDM file, including the version, date, and the originator of the file.

The TDM file header should indicate version 2.0.

The comment section of the header should include the following information: A summary of the file contents.

- Identification of the ground site in use, using the KSAT-unique identifier and location.
- Precise WGS84 and ECEF coordinates of the ground station antenna reference point (resolution to 1 mm).
- Pedestal offset distance (Lg) for the antenna
- Most recent calibration date for the TLT in use
- The common name, COSPAR ID, and the (NORAD/USSPACECOM) catalog number, if available) for the spacecraft being tracked.

If any of these items are not known, they should be indicated as "UNKNOWN".

The creation date should be the time at which the TDM file was opened (resolution of 1 second). The creation date should be the same as that used to generate the file name. The originator is given as 'KSAT'.

Required Metadata Items

This section describes metadata entries that are common across all TDM files. These are the items that are listed as "mandatory" in the [TDM spec](#). Each TDM file may have additional metadata entries, and those are

described within the specifics for each file.

TIME_SYSTEM indicates the time standard in use. Always set to UTC.

PARTICIPANT_1 indicates the ground station ID, and is one of the KSAT-unique identifiers for each of our sites. Stations that support ranging and TDM generation are listed in Appendix B.

PARTICIPANT_2 indicates the spacecraft being tracked by the antenna. Usually, this should be the COSPAR ID, but could also be the 5 digit or 9 digit satellite catalog ID.

Detailed TDM Format Definitions

For a specific description of each type of TDM file provided, see the following links:

- [TRACK](#): Radiometric tracking data (range, carrier phase, range rate):
- [ANGLE](#): Antenna pointing angles
- [SIGMET](#): Signal metrics
- [METEO](#): Meteorology